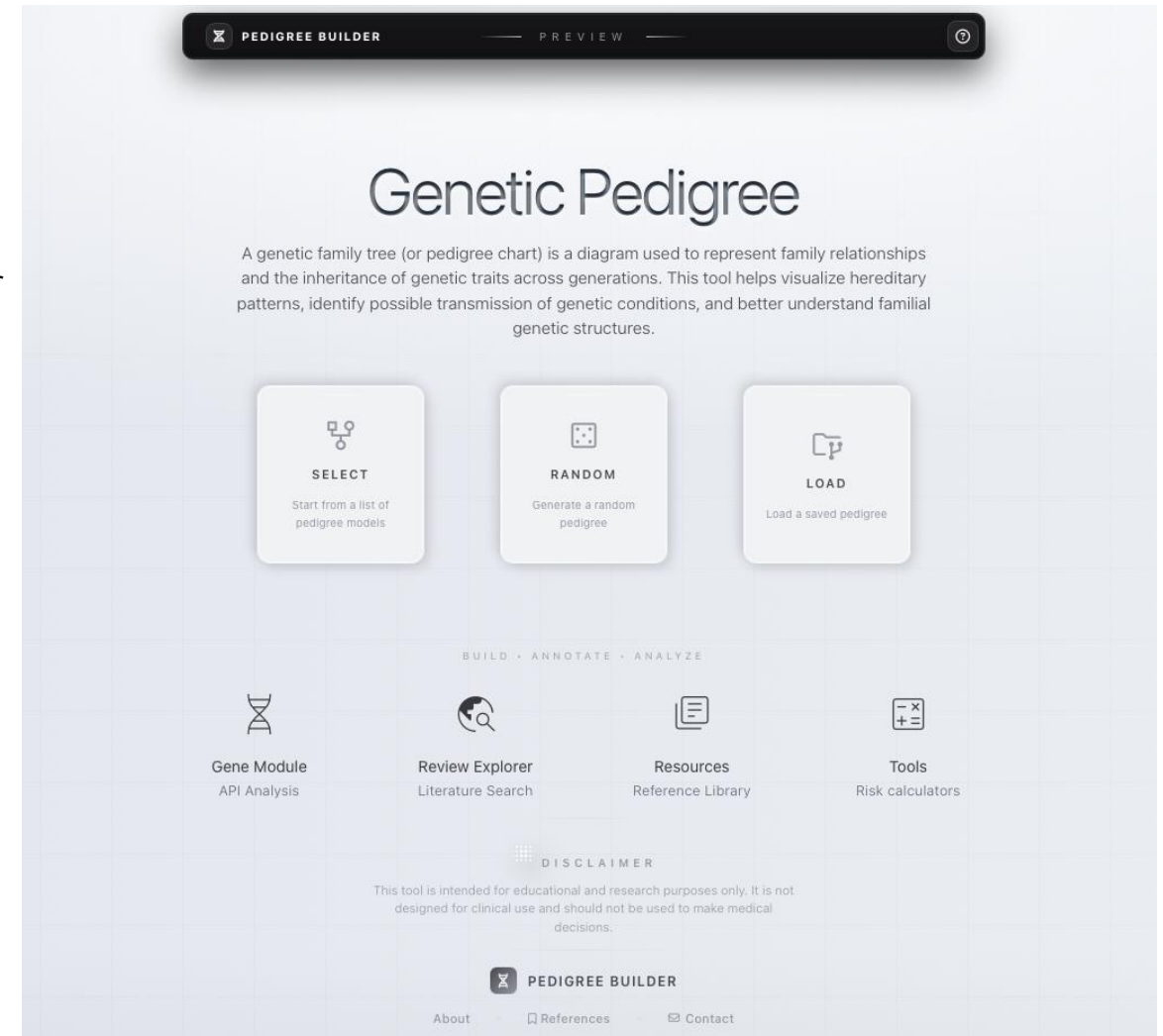


User Interface Redesign

- Redesigned the interface for a cleaner and more intuitive user experience.
- Added prominent buttons for the application's primary function: Pedigree Creation.
- Organized secondary features into dedicated Learning & Research Tools sections.
- Integrated scientific references and supporting resources.
- Improved navigation, accessibility, and overall usability.



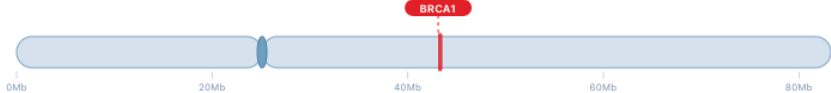
Gene Explorer

Provides technical gene information from trusted reference databases through integrated APIs, supporting research activities and molecular genetics education.

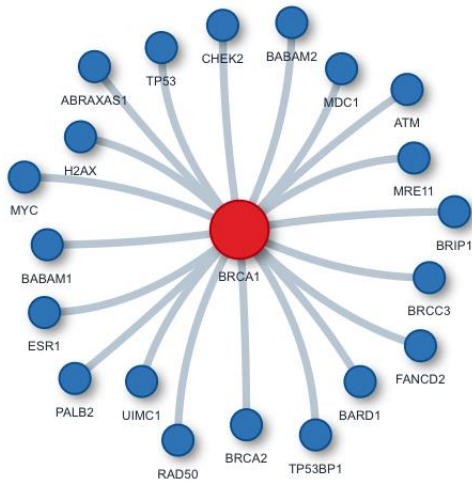
Genomic Location

Chromosome	17
Start	43,044,292
End	43,170,245
Strand	Reverse (-)
Size	125,953 bp
Biotype	protein_coding

Chromosome 17
BRCA1 (43,044,292 - 43,170,245, 125,953 bp, strand -)



[View on Ensembl](#) [View on UCSC](#)



Q Search

Overview

Details

Clinical

Pathways

Network

Literature

BRCA1 — Breast cancer type 1 susceptibility protein

P38398

P38398 1863 aa Reviewed

SYMBOL BRCA1	ENTREZ ID 672	ENSEMBL ENSG00000012048	UNIPROT P38398	CHROMOSOME 17	BIOTYPE protein-coding	HCNC 1100
OMIM 113705						

Functional Summary

This gene encodes a 190 kD nuclear phosphoprotein that plays a role in maintaining genomic stability, and it also acts as a tumor suppressor. The BRCA1 gene contains 22 exons spanning about 110 kb of DNA. The encoded protein combines with other tumor suppressors, DNA damage sensors, and signal transducers to form a large multi-subunit protein complex known as the BRCA1-associated genome surveillance complex (BASC). This gene product associates with RNA polymerase II, and through the C-terminal domain, also interacts with histone deacetylase complexes. This protein thus plays a role in transcription, DNA repair of double-stranded breaks, and recombination. Mutations in this gene are responsible for approximately 40% of inherited breast cancers and more than 80% of inherited breast and ovarian cancers. Alternative splicing plays a role in modulating the subcellular localization and physiological function of this gene. Many alternatively spliced transcript variants, some of which are disease-associated mutations, have been described for this gene, but the full-length natures of only some of these variants has been described. A related pseudogene, which is also located on chromosome 17, has been identified. [provided by RefSeq, May 2020].

ALIASES & SYNONYMS

BRCA1 BRCC1 BROVCA1 FANCS IRIS PNCA4 PPP1R53 PSCP RNF53

Publications — BRCA1 20 articles

TITLE	AUTHORS	JOURNAL	YEAR	LINK
Breast cancer gene-1 (BRCA1) potentiates maladaptive repair after kidney injury.	Ajay AK, Akinfolarin AA, Gifford CC	The Journal of experimental medicine	2025	40152784
Suppression of BRCA1 facilitates kidney regeneration.	Church KA, Zhou X, Kalluri R	The Journal of experimental medicine	2025	40152785
HSP90 buffers deleterious genetic variations in BRCA1.	Gracia B, Zhang XH, Montes P	Molecular cell	2025	41265458
BRCA1-Mediated Dual Regulation of Ferroptosis Exposes a Vulnerability to GPX4 and PARP Co-Inhibition in BRCA1-Deficient Cancers.	Lei G, Mao C, Horbath AD	Cancer discovery	2024	38552003

Review Explorer

Provides clinically relevant information from trusted medical databases through integrated APIs, including diagnostic criteria, patient management guidelines, risk assessment, and predictive medicine insights.



Diagnosis

SUGGESTIVE FINDINGS

BRCA1- and *BRCA2*-associated hereditary breast and ovarian cancer (HBOC) **should be suspected** in individuals with a personal or family history (first-, second-, or third-degree relative in either lineage) of any of the following:

- Breast cancer diagnosed at or before age 50 years
- Ovarian cancer
- Multiple (i.e., >1) primary breast cancers in either one or both breasts
- Male breast cancer
- Triple-negative (estrogen receptor-negative, progesterone receptor-negative, and human epidermal growth factor receptor 2-negative) breast cancer
- The combination of pancreatic cancer and/or prostate cancer (metastatic or Gleason score ≥7) with breast cancer and/or ovarian cancer
- Breast cancer diagnosed at any age in an individual of Ashkenazi Jewish ancestry
- Two or more relatives with breast cancer, one diagnosed at or before age 50 years
- Three or more relatives with breast cancer at any age
- A family member with a known *BRCA1* or *BRCA2* pathogenic variant

Note: (1) "Breast cancer" includes both invasive cancer and ductal carcinoma in situ. (2) "Ovarian cancer" includes epithelial ovarian cancer, fallopian tube cancer, and primary peritoneal cancer.

PROBABILITY MODELS FOR *BRCA1* AND *BRCA2* PATHOGENIC VARIANTS

Several models have been developed to estimate the likelihood that an individual or family has a germline pathogenic variant in *BRCA1* or *BRCA2* [Parmigiani et al 1998, Frank et al 2002, Antoniou et al 2004, Evans et al 2004, Tyrer et al 2004]. Even among experienced providers, the use of probability models has been shown to increase the ability to identify which individuals are more likely to have a *BRCA1* or *BRCA2* pathogenic variant [Euhus et al 2002, de la Hoya et al 2003]. For more information about probability models for *BRCA1* and *BRCA2* pathogenic variants, click [here](#).

ESTABLISHING THE DIAGNOSIS

Clinical Characteristics

Diagnosis

Management

Genetic Counseling

Molecular Genetics

Clinical Characteristics

CLINICAL DESCRIPTION

BRCA1- and *BRCA2*-associated hereditary breast and ovarian cancer (HBOC) is characterized by an increased risk for male and female breast cancer, ovarian cancer (including fallopian tube and primary peritoneal cancers), and to a lesser extent other cancers such as prostate, pancreatic, and melanoma, primarily in individuals with a *BRCA2* pathogenic variant. Estimates of malignancy risk vary considerably depending on the context in which they were derived. Table 2 is a summary of the risk for malignancy in an individual with a germline *BRCA1* or *BRCA2* pathogenic variant.

Risk of Malignancy in Individuals with a Germline *BRCA1* or *BRCA2* Pathogenic Variant

CANCER TYPE	GENERAL POPULATION RISK	RISK FOR MALIGNANCY	
		<i>BRCA1</i>	<i>BRCA2</i>
Breast	12%	55%-72% by age 70	45%-69%
Contralateral breast cancer	2% w/in 5 yrs	20%-30% w/in 10 yrs; 40%-50% w/in 20 yrs	
Ovarian	1%-2%	39%-44%	11%-17%
Male breast	0.1%	1%-2%	6%-8%
Prostate	6% by age 69 yrs	21% by age 75 yrs; 29% by age 85 yrs	27% by age 75 yrs; 60% by age 85 yrs
Pancreatic	0.5%	1%-3%	3%-5% by age 70 yrs
Melanoma (cutaneous & ocular)	1.6%	Elevated risk	

Resources

Provides access to trusted reference websites and professional databases commonly used by healthcare professionals, genetic counselors, and researchers for clinical practice and education.

Resources

External databases and tools for genomics research



Genomics

DECIPHER

Interactive database for the interpretation of genomic variants. Maps chromosomal imbalances and links them to phenotypic data.

[Learn more →](#)



Genomics

gnomAD

Genome Aggregation Database. Aggregates and harmonizes exome and genome sequencing data to provide allele frequencies across populations.

[Learn more →](#)



Literature

PubMed

Comprehensive biomedical literature database from the National Library of Medicine. Access millions of citations and abstracts.

[Learn more →](#)



Clinical

MedlinePlus

Trusted health information from the National Library of Medicine. Provides patient-friendly descriptions of conditions, genes, and treatments.

[Learn more →](#)



Clinical

Orphanet

Portal for rare diseases and orphan drugs. Provides expert-reviewed information on rare disease nomenclature, prevalence, and care guidelines.

[Learn more →](#)



Risk

CanRisk

Cancer risk prediction tool based on validated models. Calculates breast and ovarian cancer risk using genetic and family history data.

[Learn more →](#)



Tools

QuickPed

Interactive pedigree drawing tool. Quickly create and export pedigree diagrams with an intuitive drag-and-drop interface.

[Learn more →](#)



Genomics

UCSC Genome Browser

Genome visualization platform. Browse and annotate genomic data with tracks for genes, variants, conservation, and regulatory elements.

[Learn more →](#)



Education

NHGRI Educational Resources

Educational materials from the National Human Genome Research Institute. Fact sheets, glossaries, and multimedia resources about genomics.

[Learn more →](#)

Punnett Square

Interactive genotype and phenotype explorer

CROSS SETUP

Transmission mode

Autosomal recessive

Number of traits

1

Cell display

☒ Genotype ☐ Phenotype

☒ Show frequency tables

Autosomal modes support multiple traits. Choose a causal allele for each trait.

TRANSMISSION SETTINGS

Trait 1 causal / risk allele

A

PARENT ALLELES

Mother / Parent 1

A1

A

A2

a

Father / Parent 2

A1

A

A2

a

Reset

Export CSV

Punnett Square

An interactive learning tool that helps users understand inheritance patterns by simulating different modes of genetic transmission and visualizing genotype and phenotype outcomes.

MOST COMMON OUTCOMES

- T1 Carrier 50.0%
- T1 Affected 25.0%
- T1 Unaffected 25.0%

	A	a
A	AA	Aa
a	Aa	aa

Genotype frequencies

Ratio: 2 : 1 : 1

Value	Count	Percent
☐ Aa	2	50.00%
☐ aa	1	25.00%
☐ AA	1	25.00%

Phenotype frequencies

Ratio: 2 : 1 : 1

Value	Count	Percent
T1 Carrier	2	50.00%
T1 Affected	1	25.00%
T1 Unaffected	1	25.00%

Punnett Square

Interactive genotype and phenotype explorer

- Classic Punnett
- Mixed per trait
- Autosomal dominant
- ☒ Autosomal recessive
- X-linked dominant
- X-linked recessive
- Y-linked
- Mitochondrial

Enhanced Pedigree Display

- More comprehensive pedigree visualization.
- Added statistics and analytical indicators.
- Expanded legend customization options.
- Support for overlaying multiple traits and characteristics.
- Improved readability for complex pedigrees.

Home

Enter pedigree title...

🔄

🗑️

Legend / Options

GENDER

Male

Female

Undefined

ASSIGNED AT BIRTH

AFAB Assigned Female at Birth

AMAB Assigned Male at Birth

UAB Undetermined at Birth

STATUS MARKERS

Deceased

Miscarriage

Adopted

Proband

Infertility

Phenotypes

+ Create Phenotype

Phenotype 1
2 individuals

Phenotype 2
2 individuals

Phenotype 3
4 individuals

Phenotype 4
2 individuals

Draw Pedigree

INDIVIDUALS 9

FOUNDERS 4

NON-FOUNDERS 5

LEAVES 1

GENERATIONS 4

INBREEDING LOOPS 2

FEMALES 4

Press and hold a card to highlight individuals. Click Inbreeding Loops to inspect detected loops.

Individual

II-2

Female

Alive

🗑️

Last name

First name

DOB 11/05/2026

Age 0 Years

Deceased

Family Gold

PARENTS

Add Parents

SIBLINGS

Sister

Brother

Unknown

Twins

ADDING Twins

Twins Triplet

MZ DZ Unknown

♀ ♂ ?

+ Add

CHILDREN

Daughter

Son

Unknown

Miscarriage

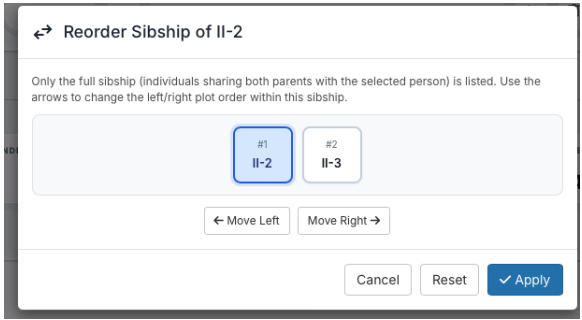
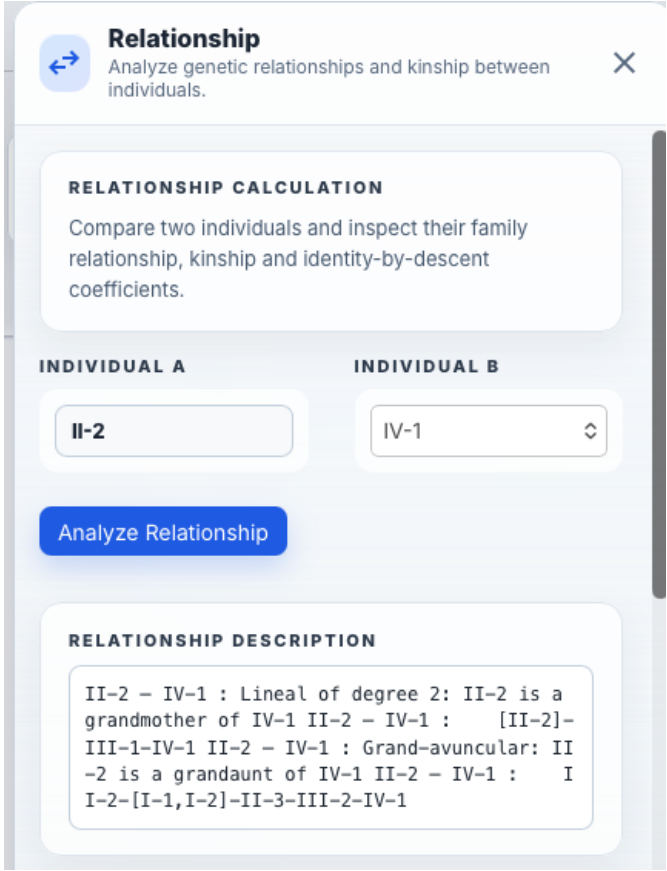
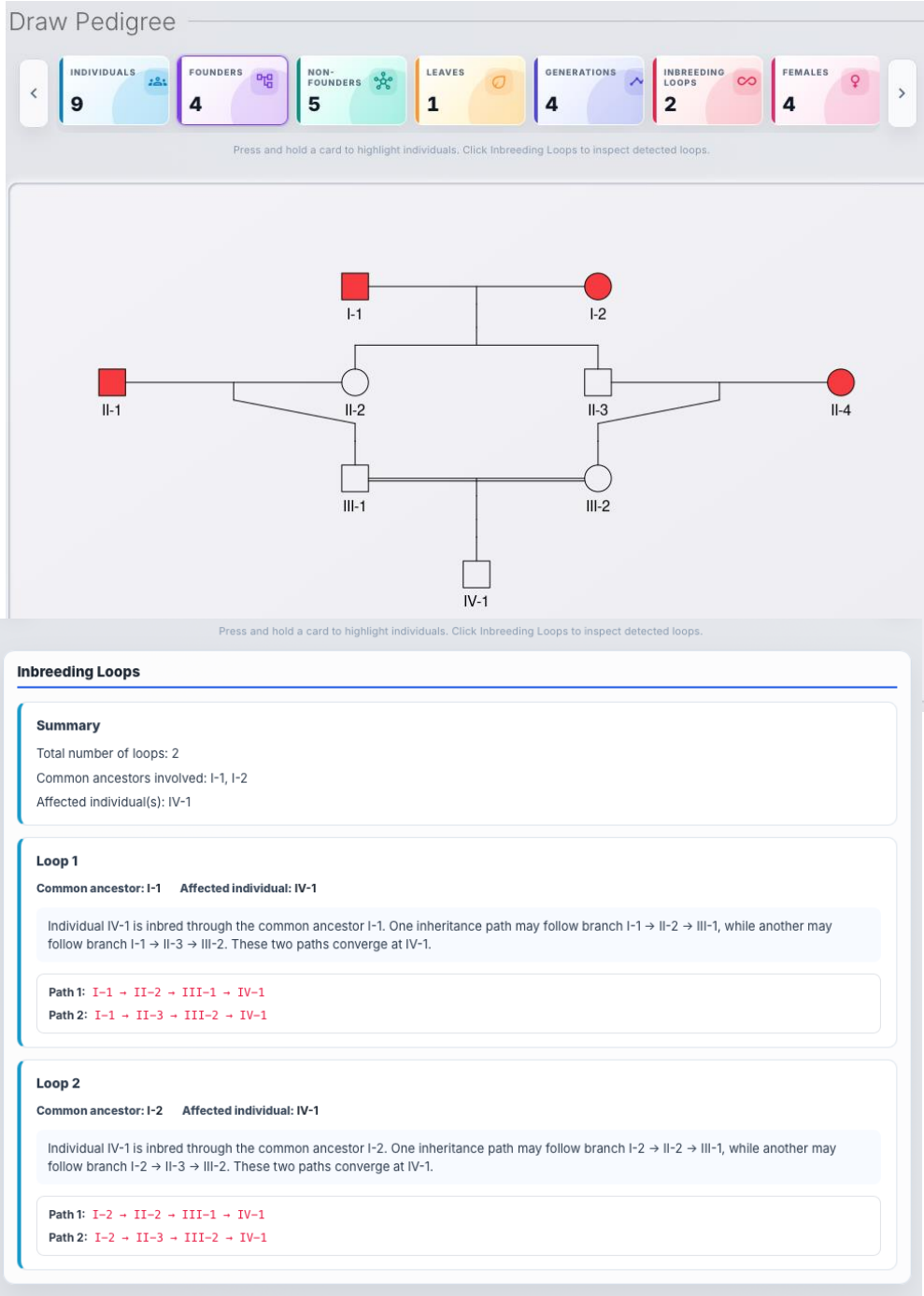
INDIVIDUAL	MOTHER	FATHER	F	STATUS	PHENOTYPES	SIBLINGS	SPOUSES	CHILDREN
I-1	—	—	0	Alive	—	—	I-2	I-3
I-2	—	—	0	Alive	—	—	I-1	I-2
II-1	—	—	0	Alive	Phenotype 4	—	II-2	II-1
II-2	I-2	I-1	0	Alive	Phenotype 3	II-3	II-1	II-1
II-3	I-2	I-1	0	Alive	—	II-2	II-4	II-2
II-4	—	—	0	Alive	—	—	II-3	II-2

Showing 1 to 6 of 9 individuals

1 2

Advanced Analysis Features

- Automated kinship and relationship calculations.
- Detection and visualization of consanguinity loops.
- Repositioning and reordering of individuals within the pedigree.
- Improved pedigree organization and interpretation.



Individual Report Generation

- Double-click on an individual to generate a structured report.
- Built on a modular block-based architecture.
- Integrates pedigree findings, clinical information, and genetic analysis.
- Designed to support clinical assessment and patient management.

The screenshot displays the 'Individual II-2 Report' in the MedNotes Pro application. The interface includes a dark sidebar on the left with a search bar and a 'PAGES' section containing 'Template Report'. Below this is a list of report sections: Chief Complaint, Medical History, Risk Factors, Clinical Exam, Follow-up, and Pedigree. The main content area shows a 'Template Report' with two bullet points: 'BP elevated at 148/92 despite Lisinopril 20mg — uptitrating + adding Amlodipine' and 'HbA1c trending up (7.2% from 6.9%) — nutrition referral placed, may need medication adjustment'. A 'ADD BLOCK' menu is open, showing options like Columns, Table, Kanban, Tabs, Sidebar, and Mindmap. A pedigree chart is visible in the background, showing a family tree with individuals labeled I-1, I-2, II-2, II-3, II-4, III-1, and III-2.

This screenshot shows the 'Medical History' section of the report. It is organized into categories: Medical, Surgical, Family, and Allergies. The 'Medical' section lists 'Hypertension (diagnosed 2015, on Lisinopril 20mg)', 'Type 2 Diabetes Mellitus (diagnosed 2018, on Metformin 1000mg BID)', and 'Dyslipidemia (on Atorvastatin 40mg)'. The 'Surgical' section lists 'Appendectomy (1985)' and 'Right knee arthroscopy (2012)'. The 'Family' section lists 'Father — MI at age 62, deceased', 'Mother — Type 2 Diabetes, Hypertension', and 'Brother — Coronary artery disease, stent placement at 58'. The 'Allergies' section lists 'Penicillin — rash' and 'Sulfa drugs — anaphylaxis'.

